Department of Mathematics and Computer Science
NATIONAL UNIVERSITY OF LESOTHO

M3503 **Tutorial 4** **October 2025**

1. Use definition of Laplace transform to find $\mathcal{L}\{f(t)\}$

   (a) $f(t) = \begin{cases} -1, & 0 \le t < 1 \\ 1, & t \ge 1 \end{cases}$

   (b) $f(t) = \begin{cases} 4, & 0 \le t < 2 \\ 0, & t \ge 2 \end{cases}$

   (c) $f(t) = \begin{cases} t, & 0 \le t < 1 \\ 1, & t \ge 1 \end{cases}$

   (d) $f(t) = \begin{cases} 2t+1, & 0 \le t < 1 \\ 0, & t \ge 1 \end{cases}$

   (e) $f(t) = \begin{cases} \sin t, & 0 \le t < \pi \\ 0, & t \ge \pi \end{cases}$

   (f) $f(t) = \begin{cases} 0, & 0 \le t < \pi/2 \\ \cos t, & t \ge \pi/2 \end{cases}$

   (g) $f(t) = e^{t+7}$

   (h) $f(t) = e^{-2t-5}$

   (i) $f(t) = te^{4t}$

   (j) $f(t) = t^2 e^{-2t}$

   (k) $f(t) = e^{-t}\sin t$

   (l) $f(t) = e^{t}\cos t$

   (m) $f(t) = t\cos t$

   (n) $f(t) = t\sin t$

2. Find $\Delta L\{f(t)\}$

   (a) $f(t) = 4t^2 - 5\sin 3t$

   (b) $f(t) = \cos 5t + \sin 2t$

   (c) $f(t) = \sin 2t \cos 2t$

   (d) $f(t) = \cos^2 t$

3. Use the fact that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ to find the Laplace transform of the given function

   (a) $f(t) = t^{-1/2}$

   (b) $f(t) = t^{1/2}$

   (c) $f(t) = t^{3/2}$

   (d) $f(t) = 2t^{1/2} + 8t^{5/2}$

4. Find the given inverse Laplace transform

   (a) $\Delta L^{-1}\left\{\dfrac{2s-4}{(s^2+s)(s^2+1)}\right\}$

   (b) $\Delta L^{-1}\left\{\dfrac{1}{s^4-9}\right\}$

   (c) $\Delta L^{-1}\left\{\dfrac{1}{(s^2+1)(s^2+4)}\right\}$

   (d) $\Delta L^{-1}\left\{\dfrac{6s+3}{s^4+5s^2+4}\right\}$

5. Use the Laplace transform to solve the given initial-value problem

   (a) $y' + 6y = e^{4t}, \quad y(0) = 2$

   (b) $y' - y = 2\cos 5t, \quad y(0) = 0$

   (c) $y'' + 5y' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 0$

   (d) $y'' - 4y' = 6e^{3t} - 3e^{-t}, \quad y(0) = 1, \quad y'(0) = -1$

   (e) $y'' + y = \sqrt{2}\sin\sqrt{2}t, \quad y(0) = 10, \quad y'(0) = 0$

   (f) $y'' + 9y = e^{t}, \quad y(0) = 0, \quad y'(0) = 0$

   (g) $2y''' + 3y'' - 3y' - 2y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1$

   (h) $y''' + 2y'' - y' - 2y = \sin 3t, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1$

   (i) $y' + y = e^{-3t}\cos 2t, \quad y(0) = 0$

   (j) $y'' - 2y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = 3$

   (k) $y'' + 4y = 10\cos 5t, \quad y(0) = 0, \quad y'(0) = 0$

(l) $y'' + 2y = 4t, \quad y(0) = 0, \quad y'(0) = 0$

6. Use the Laplace transform to solve the given initial-value problem

(a) $y'' + 2y' + y = 0, \quad y(0) = 1, \quad y'(0) = 1$

(b) $y'' - 4y' + 4y = t^3 e^{2t}, \quad y(0) = 0, \quad y'(0) = 0$

(c) $y'' - 6y' + 9y = t, \quad y(0) = 0, \quad y'(0) = 1$

(d) $y'' - 4y' + 4y = t^3, \quad y(0) = 1, \quad y'(0) = 0$

(e) $y'' - 6y' + 13y = 0, \quad y(0) = 0, \quad y'(0) = -3$

(f) $2y'' + 20y' + 51y = 0, \quad y(0) = 2, \quad y'(0) = 0$

(g) $y'' - y' = e^t \cos t, \quad y(0) = 0, \quad y'(0) = 0$

(h) $y'' - 2y' + 5y = 1 + t, \quad y(0) = 0, \quad y'(0) = 4$

(i) $y'' + 2y' + y = 0, \quad y'(0) = 2, \quad y(1) = 2$

(j) $y'' + 8y' + 20y = 0, \quad y(0) = 0, \quad y'(\pi) = 0$

(k) $y'' + 4y = f(t), \quad y(0) = 0, \quad y'(0) = -1$, where

$$f(t) = \begin{cases} 1, & 0 \le t < 1 \\ 0, & t \ge 1 \end{cases}$$

$y'' + 4y = \sin t \Delta U(t - 2\pi), \quad y(0) = 1, \quad y'(0) = 0$

(l) $y'' - 5y' + 6y = \Delta U(t - 1), \quad y(0) = 0, \quad y'(0) = 1$

(m) $y'' + 4y' + 3y = 1 - \Delta U(t - 2) - \Delta U(t - 4) + \Delta U(t - 6), \; y(0) = 0, \quad y'(0) = 0$

7. Use the Laplace transform to solve the given integral equation or integrodifferential equation

(a) $f(t) + \int_0^t (t - \tau) f(\tau) d\tau = t$

(b) $f(t) = 2t - 4 \int_0^t \sin \tau f(t - \tau) d\tau$

(c) $f(t) = te^t + \int_0^t \tau f(t - \tau) d\tau$

(d) $f(t) + 2 \int_0^t f(\tau) \cos(t - \tau) d\tau = 4e^{-t} + \sin t$

(e) $f(t) + \int_0^t f(\tau) d\tau = 1$

(f) $f(t) = \cos t + \int_0^t e^{-\tau} f(t - \tau) d\tau$

(g) $f(t) = 1 + t - \frac{8}{3} \int_0^t (\tau - t)^3 f(\tau) d\tau$

(h) $t - 2f(t) = \int_0^t \left(e^{\tau} - e^{-\tau}\right) f(t - \tau) d\tau$

(i) $y'(t) = 1 - \sin t - \int_0^t y(\tau) d\tau, \quad y(0) = 0$

8. (a) Laguerre's differential equation

$$ty'' + (1 - t)y' + ny = 0$$

is known to possess polynomial solutions when n is a nonnegative integer. These solutions are naturally called Laguerre polynomials and are denoted by $L_n(t)$. Find $y = L_n(t)$, for $n = 0, 1, 2, 3, 4$ if it is known that $L_n(0) = 1$.

(b) Show that

$$\Delta L \left\{ \frac{e^t}{n!} \frac{d^n}{dt^n} t^n e^{-t} \right\} = Y(s),$$